

Agastya Sharma

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, C++

Frameworks and Libraries: React, Next.js, Redux, Tailwind CSS, React Native, Node.js, Flask, FastAPI, PyTorch

DevOps and Cloud: Docker, Kubernetes, AWS, CI/CD, GitHub Actions, Redis, Vercel

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Arts in Computer Science, Co-op (GPA: 3.90 / 4.00)

2025 – 2030

- President's Scholarship of Distinction; Michael Egan Entrance Scholarship in Computer Science

WORK EXPERIENCE

Sylphia Consulting Inc.

Sep 2025 – Present

Full Stack Developer

Toronto, ON

- Led a team of 5 interns using Agile sprint methodology to deliver 3 production-ready applications within 4 months
- Developed a full-stack web application using Next.js, Redux, Node.js, and PostgreSQL for nutrition analysis
- Integrated RESTful APIs using Express.js and FastAPI, handling over 10,000 requests with sub-200ms latency
- Built CI/CD pipelines with GitHub Actions to automate testing, reducing test execution time by 50 percent
- Containerized and deployed microservices using Docker and Kubernetes on AWS
- Trained and deployed custom machine learning models for computer vision and feature recognition
- Implemented asynchronous background task processing using Redis for high-volume workloads

Steel City Codes Ontario

Jun 2022 – Jul 2024

Founder and Technical Curriculum Lead

Brampton, ON

- Scaled a coding education program to over 100 students across two terms with an 82 percent retention rate
- Designed technical curricula and assignments in Python, HTML, CSS, and JavaScript
- Delivered hands-on technical workshops, simplifying complex software engineering concepts for diverse audiences

FEATURED PROJECTS

Nutrition Application — Next.js, TypeScript, ShadCN, PyTorch PostgreSQL

Jan 2026

- Created a full-stack Next.js web application using for calorie tracking with personalized insights
- Developed a custom CNNs in PyTorch for multi-dish image classification and depth estimation
- Built OAuth 2.0-secured RESTful APIs using Node.js for uploads, analytics, and PostgreSQL data management
- Integrated multiple external APIs for real-time nutritional data and generate dynamic suggestions using Redis

Music Insights Dashboard — Flask, Pandas, Jinja, JavaScript, Bootstrap, SQLite

Sep 2025

- Built a full-stack web application providing personalized Spotify analytics and music recommendations
- Implemented OAuth 2.0 authentication to access user listening history and library data
- Collected and processed over 1,000 pages of music data using BeautifulSoup and Selenium
- Developed a content-based recommendation engine using audio feature similarity scoring in Pandas

Oil Price Prediction Model — Python, Pandas, NumPy, Scikit-learn, PyTorch

Jan 2025

- Developed an end-to-end machine learning pipeline to forecast WTI crude oil prices using time-series data
- Implemented hybrid CNN-LSTM and GRU neural network architectures in PyTorch
- Evaluated model accuracy using RMSE and MAE metrics and performed hyperparameter tuning

3D ASCII Renderer — Python, NumPy, Linear Algebra

Dec 2024

- Built a real-time 3D rendering engine projecting parametric surfaces as ASCII art
- Applied linear algebra transformations and depth buffering for perspective rendering
- Calculated lighting intensity using surface normals and vector dot products

AWARDS

Incubator Hacks Winner; International Diamond Challenge State Qualifiers; Two-Time Provincial and Regional DECA Champion in ENT and ETDM; PGCC Continental Finalist; Tech Under 20 Winner; Top 25% CSMC, Euclid, and CCC